

Linear Algebra — Exercises

Eigenvalues and eigenvectors

Exercise 1. Find the eigenvalues of $A = M_S(F)$:

(a) $A = \begin{pmatrix} 2+i & 1 \\ 2 & 2-i \end{pmatrix}$

(b) $A = \begin{pmatrix} 2 & -1 & 2 \\ 5 & -3 & 3 \\ -1 & 0 & -2 \end{pmatrix}$

Exercise 2. Find all eigenvalues and a basis of each eigenspace for the following linear operators:

(a) $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $F(x, y) = (x, x + y)$

(b) $F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $F(x, y, z) = (2x + y, y - z, 2y + 4z)$

(c) $F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $F(x, y, z) = (x - z, 2y, x + z)$

(d) $F: \mathbb{C}^2 \rightarrow \mathbb{C}^2$, $F(x, y) = (-y, x)$

(e) $F: \mathbb{C}^3 \rightarrow \mathbb{C}^3$, $F(x, y, z) = (x - z, 2y, x + z)$.

Exercise 3. Assuming that each of the following matrices represents a linear operator F in a basis S , find all eigenvectors and a basis of each eigenspace:

(a) $A = M_S(F) = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 3 \end{pmatrix}$

(b) $A = M_S(F) = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & -1 \\ -1 & 1 & 4 \end{pmatrix}$

(c) $A = M_S(F) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Exercise 4. For each operator F from Exercise 3, determine whether there exists a basis T such that the matrix of the operator F in the basis T is diagonal.

Exercise 5. For the matrix $A = \begin{pmatrix} 1 & 2 & -1 \\ 1 & -1 & 0 \\ 2 & -2 & 0 \end{pmatrix}$ calculate A^{100} .

Exercise 6. For a linear mapping $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $F(1, 2) = (2, 4)$ and $F(-2, 1) = (-2, 1)$, calculate $F^{150}(1, 0)$.